The Decomposition of Intrahousehold Gender Gaps in Agricultural Technology Adoption

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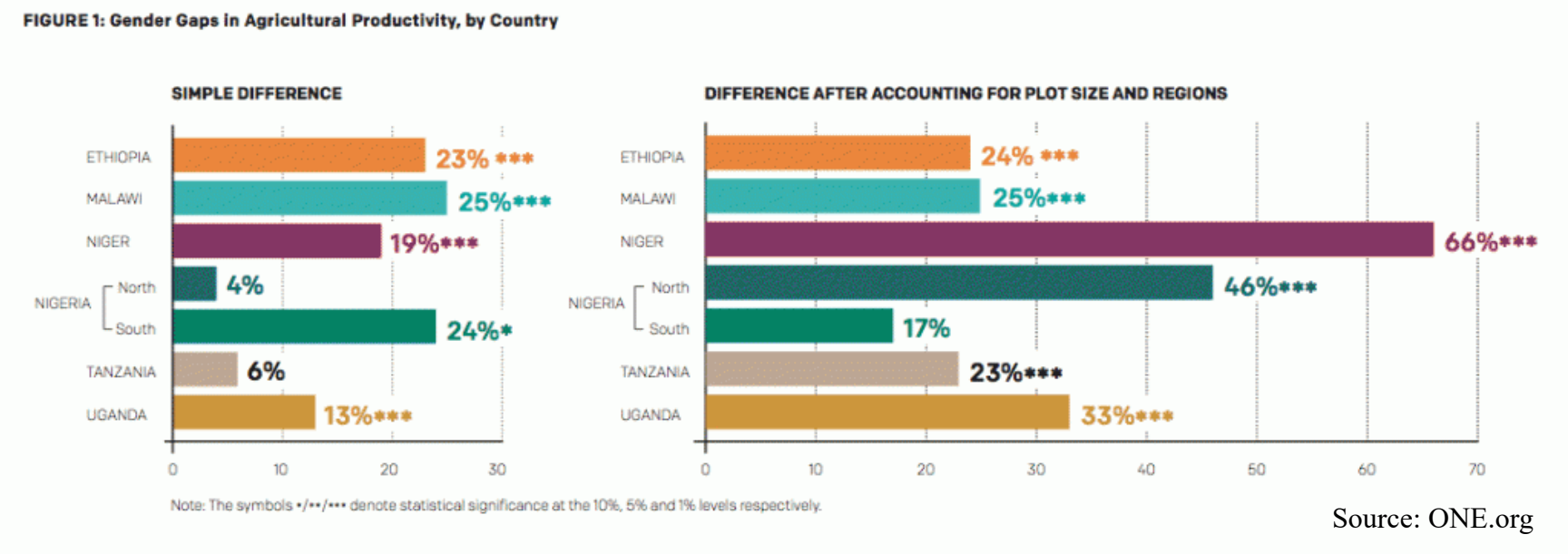
Gender and Agriculture

- ✓ Gender impacts agricultural productivity
- ✓ Spousal disparities within households affect decision-making.



Source: ONE.org

Measurement Errors in Agricultural Data

Challenges of farmer-reported production data:

- ✓ Farmer perception vs. actual production
- ✓ Inaccuracies in self-reported data

Impacts of measurement errors on understanding gender dynamics

- ✓ Misrepresentation of gender roles.
- ✓ Underestimation of gender disparities.
- ✓ Reliance on self-reported survey data with potential measurement errors.
- ✓ Focus on household head data, neglecting spousal disparities.

Sampling and sex-disaggregation of cassava-related technology adoption in Nigeria

Author	Sample Size	Unit of Analysis	Sex-Disaggregation	Gender-Related Results
Awotide et al., 2014	952	Household head	89% male respondent 11% female respondent 89% married	**Gender not significant in ICV adoption**
Nwakor et al., 2011	150	Farmer	45.3% male respondent 54.7% female respondent 70.7% married	**Women are more likely to adopt**
Obisesan, 2014	482	Household	76.23% male respondent 23.8% female respondent	**Men are more likely to adopt**
Udensi et al., 2011	510	Farmer	43.5% male respondent 56.5% female respondent 78.8% married	**Gender is not significant in ICV adoption**
Wossen et al., 2017	2500	Household head	89% male respondent 11% female respondent 30% spouses	**Men are more likely to adopt ICV**; **Gender is not significant** when DNA fingerprinting tests measure adoption.

Research Gap

How do spousal disparities lead to measurement errors and a biased understanding of technology adoption?

How can we improve methodology to capture gender dynamics in technology adoption?

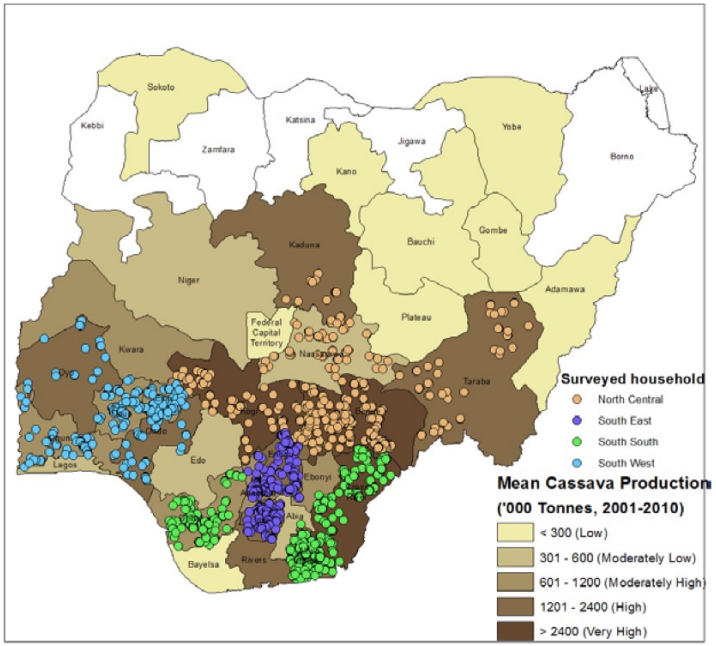
Case Study: Agricultural Production in Nigeria

Assessing gender dynamics:

- ✓ Improved crop varieties can boost productivity
- ✓ Household roles impact technology adoption Challenges for women
- ✓ Women's pivotal role in cassava cultivation
- ✓ Integrating intrahousehold data with DNA fingerprinting tests of **improved cassava variety (ICV)** adoption at the household level.

Household cassava dataset in Nigeria

Category	Variable	Husbands N	Husbands %	Wives N	Wives %
Marital Status	Married, living with spouse/s	729	100.0%	729	100.0%
Relationship to Household	Head	726	99.6%	13	1.8%
	Spouse	3	0.4%	716	98.2%
Main Occupation	Farmer (crop and livestock)	588	80.7%	490	67.2%
	Self-employed off farm	81	11.1%	160	21.9%
	Other	60	8.2%	79	10.9%
Zone	1 (North)	186	25.5%	186	25.5%
	2 (South-South)	238	32.6%	238	32.6%
	3 (Southeast)	165	22.6%	165	22.6%
	4 (Southwest)	140	19.2%	140	19.2%



Wossen, Tesfamicheal, et al. "The cassava monitoring survey in Nigeria." IITA. Ibadan: Nigeria 66 (2017).

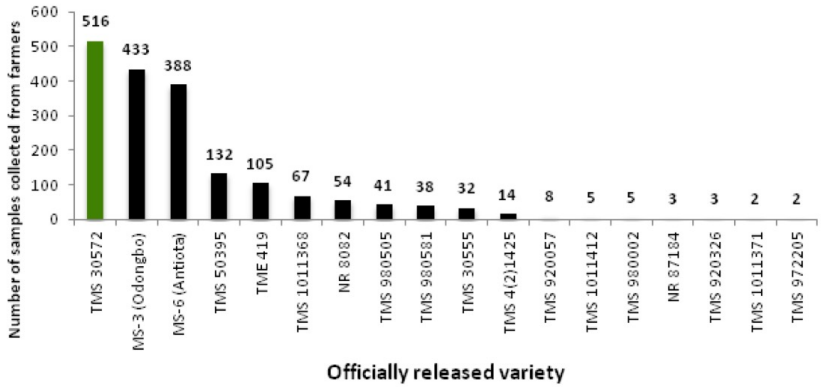
DNA fingerprinting dataset in Nigeria

Percentage of improved and/or released varieties on the farmer's field based on DNA fingerprinting.

Varieties	GG	LR	Total	%	Cumulative %
Improved AND released	919	3	922	12.50	12.50
Improved NOT released	267	491	758	10.28	22.78
Single accessions in GG cluster	110		110	1.49	24.27
Not in Library, in GG cluster	598		598	8.11	32.38
Matching LR, in GG cluster	55		55	0.75	33.12
Released but TMEB1, TMEB2..		926	926	12.55	45.68
Local varieties		4007	4007	54.32	54.32
Total	1949	5427	7376	100.00	

GG=genetic gain; LR=landrace;

Number of samples matching released varieties



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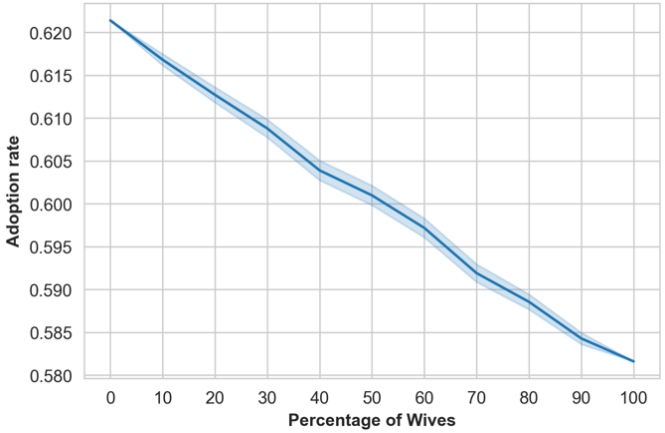
Empirical framework

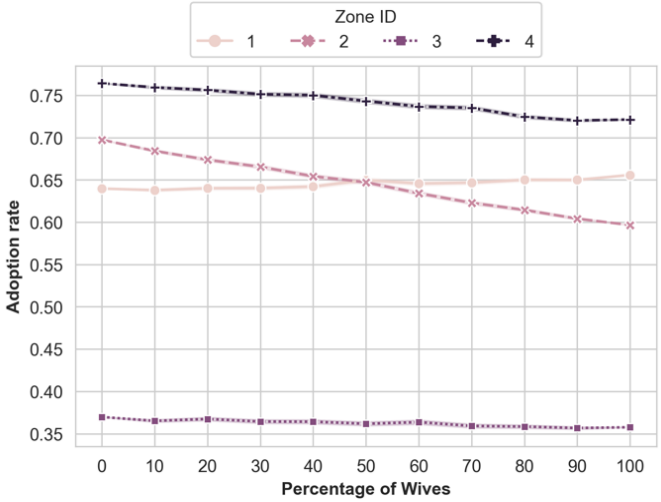
1. Representing **household adoption behavior** based on self-reported data through Monte Carlo simulations.
2. **Potential biases** in regression analysis resulting from the oversight of spousal discrepancies in self-reported data.
3. Blinder-Oaxaca decomposition to dissect the **underlying mechanisms contributing** to these disparities between spouses.

Household adoption rate analysis

Monte Carlo simulation - 500 iterations



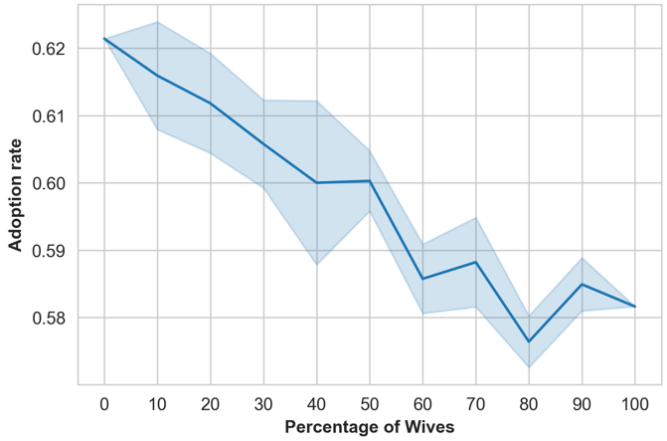
a. Population



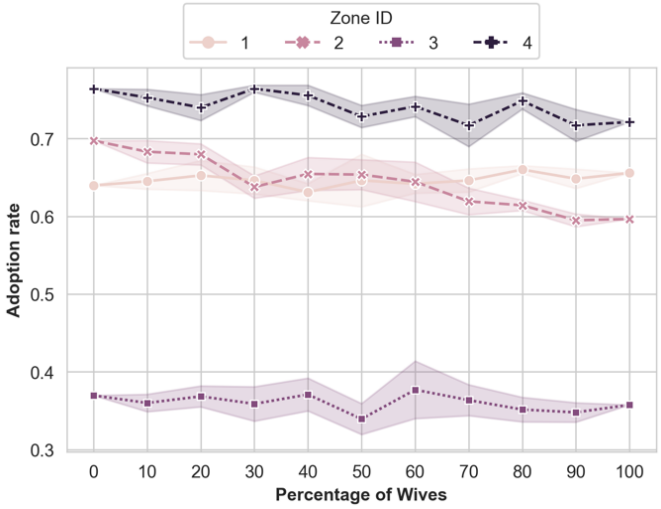
b. By region

Household adoption rate analysis

Monte Carlo simulation - 5 iterations



a. Population



b. By region

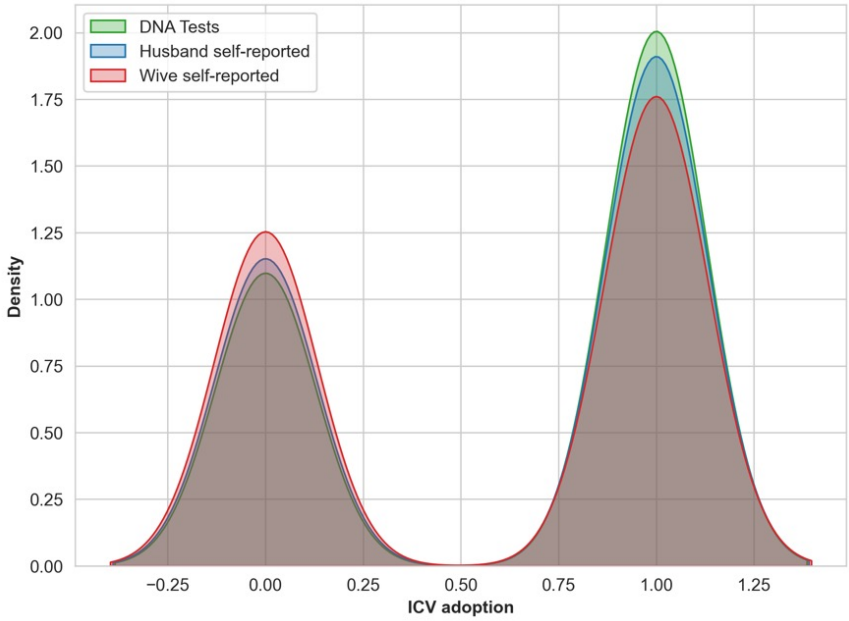
Mean differences by gender

	DNA	Husbands	Wives	Paired t-test		
	[1]	[2]	[3]	[1]-[2]	[1]-[3]	[2]-[3]
ICV adoption (1: have adopted ICVs; 0: otherwise)	0.645	0.621	0.582	0.977	2.598***	2.224**
Managed plots						
Number of ICV plots managed by husbands		0.978				
Number of ICV plots managed by wives			0.276			
Joint Managed		1.414	1.656			
Membership of a coop						
Household (reported by household heads)		0.262	0.296			
Husband Only		0.067				
Wife Only			0.029			
Credit Access						
Household (reported by household heads)		0.451	0.379			
Husband Only		0.112				
Wife Only			0.033			
Extension Access						
Household (reported by household heads)		0.444	0.287			
Husband Only		0.222				
Wife Only			0.008			
Age		50.543	40.435			
Education		9.342	8.029			

Self-reported adoption vs DNA fingerprinted data

For the self-reported adoption of ICV by females, the probability of a **negative response (no adoption)** is **higher** compared to the results obtained from DNA testing.

Highlight the occurrence of **false negatives** in the female self-reported data.



Variable	T-Statistic	P-Value
Female self-reported	**-2.508**	0.012
Male self-reported	-0.870	0.384
Female self-reported (Household head)	-0.433	0.673

Non-classical measurement error (NCME)

- ✓ Systematic spousal disparities (i.e., husbands/wives systematically overreported/underreported adoption).
- ✓ Estimates are attenuated when one gender systematically underreports the adoption.

$$\widehat{y_i} = x_i'\beta + \epsilon_i$$

where:

$\widehat{y_i}$ represents the observed adoption behavior reported in survey data,

x_i is a matrix of explanatory variables (socio-economic, farm management)

β is the coefficient vector estimated by the regression model

ϵ_i is the error term.

Non-classical measurement error (NCME)

$y_i = 1$ for adoption and $y_i = 0$ for non-adoption

Probit/Logistic model:

$$y_i = 1 = F(x_i'\beta) \qquad y_i = 0 = 1 - F(x_i'\beta)$$

Probabilities of false positives (α_{0i}) and false negatives (α_{1i}):

$$\mathrm{P}(y_i = 1) = \mathrm{P}(\widehat{y_i} \leq 0) = \alpha_{0i} \qquad \mathrm{P}(y_i = 0) = \mathrm{P}(\widehat{y_i} > 0) = \alpha_{1i}$$

The estimated marginal effects based on the observed data:

$$\frac{\partial \mathrm{P}(y = 1|x)}{\partial x} = \alpha_0 + (1 - \alpha_0 - \alpha_1)F(x_i'\beta)$$

Regression results (Probit model)

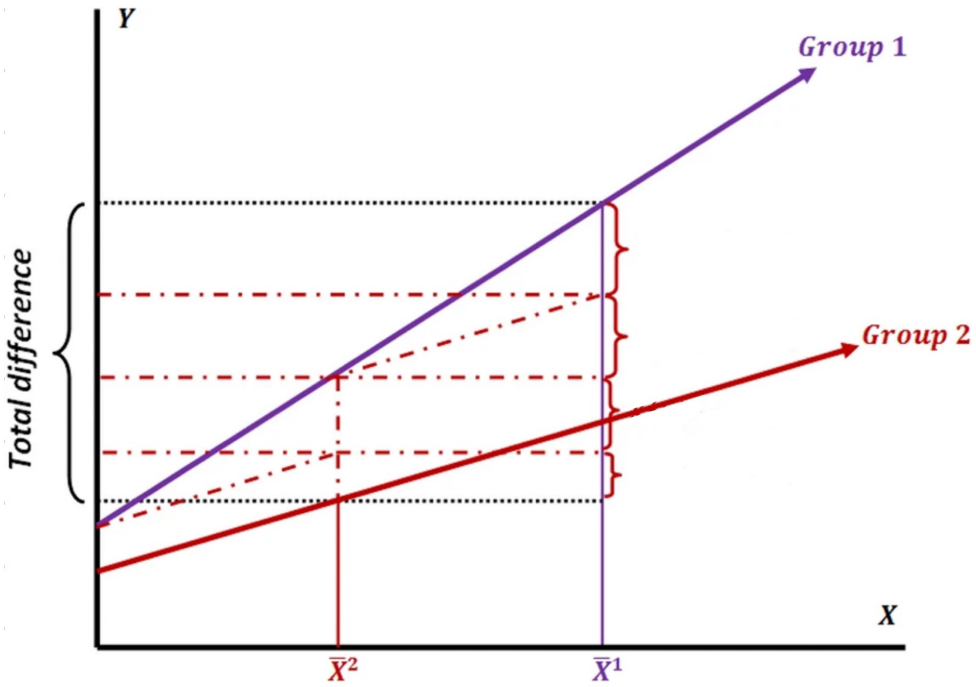
	Husbands		Wives	
	Self-reported	DNA	Self-reported	DNA
Joint Coop	**0.130****	-0.009	**0.076**	-0.062
	(0.063)	(0.179)	(0.112)	(0.059)
Household Credit	-0.040	-0.025	-0.008	0.103
	(0.065)	(0.122)	(0.102)	(0.142)
Extension Access	**0.432*****	**0.232****	0.079	-0.111
	(0.064)	(0.096)	(0.113)	(0.137)
Consumption (%)	**-0.002**	-0.001	**-0.004*****	0.003
	(0.004)	(0.007)	(0.001)	(0.002)
N	729	729	729	729

1. The dependent variables are the self-reported ICV adoption and DNA fingerprinting.
2. (*, **, ***) indicate statistical significance at the 10%, 5%, and 1% level.
3. Standard errors are clustered by region and reported in parentheses.
4. Controls for regions, Age, Education, Household size, and Main occupation.

Blinder-Oaxaca decomposition

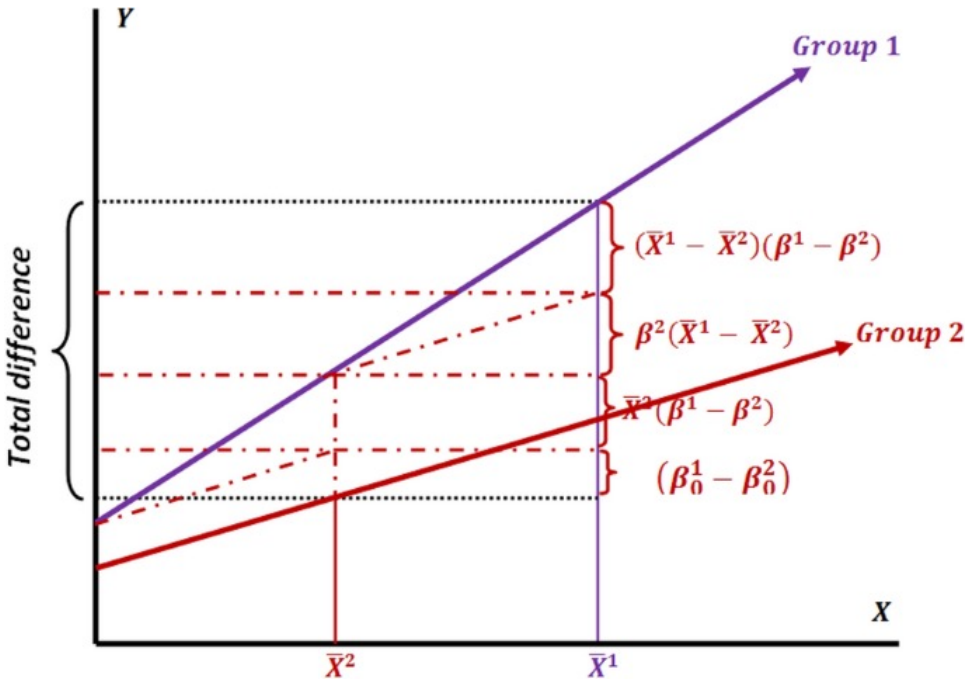
- ✓ Decomposition technique for group disparities.
- ✓ Explain the difference in the **means** of a dependent variable between **two groups.**
- ✓ Decompose the gap into **within-group** and **between-group**.

$$\Delta\bar{Y} = \bar{Y}^1 - \bar{Y}^2$$



Blinder-Oaxaca decomposition

$$
\begin{aligned}
\Delta\bar{Y} &= \bar{Y}^1 - Y^2 \\
&= \bar{X}^1\beta^1 - \bar{X}^2\beta^2 \\
&= \cdots \\
&= \beta^2(\bar{X}^1 - \bar{X}^2) + \\
&\quad \bar{X}^2(\beta^1 - \beta^2) + \\
&\quad (\bar{X}^1 - \bar{X}^2)(\beta^1 - \beta^2) \\
&= Endowment + \\
&\quad Coefficients + \\
&\quad Interaction
\end{aligned}
$$



Rahimi, Ebrahim, and Seyed Saeed Hashemi Nazari. "A detailed explanation and graphical representation of the Blinder-Oaxaca decomposition method with its application in health inequalities." Emerging Themes in Epidemiology 18.1 (2021): 12.

The Blinder-Oaxaca decomposition results

	Wife plot manager	Husband plot manager	Gap
Mean by gender	**0.167**	**0.647**	-0.480
	Endowments	Coefficients	Interaction
Aggregate decomposition	-0.019	**-0.426 *****	-0.035
	(0.046)	(0.058)	(0.055)
Share of the gap	3.96\%	88.75\%	7.29\%
Detailed decomposition			
Joint Coop	0.000	-0.029	-0.000
	(0.004)	(0.025)	(0.004)
Household Credit	0.005	0.018	-0.003
	(30.007)	(0.046)	(0.008)
Extension Access	**-0.032 ****	**-0.094 ****	0.033 *
	(0.016)	(0.047)	(0.018)
Consumption (%)	0.004	0.136	-0.004
	(0.004)	(0.091)	(0.005)

Conclusions

- ✓ Intrahousehold disparities in agricultural technology adoption and mechanisms of gender inequalities in ICV adoption.
- ✓ Offers improved methodology for analyzing gender gaps in technology adoption.
- ✓ Data best practices around intentionally including more women as respondents regardless of their household headship status.
- ✓ Systematic differences in access to agricultural education, extension, and financial resources contribute to a lower rate of ICV adoption between spouses.

Thank you